Linear combination of vectors

In the space $\mathbb{R}^2$, consider the vector $(4,2)$. It can be expressed as $(4,2) = 4(1,0)+2(0,1)$. The geometric meaning of this linear combination is that the coefficients tell us how much should we scale the vectors $(1,0)$ and $(0,1)$ and then complete the parallelogram so that we get the vector $(4,2)$ via vector addition.

We can also express $(4,2) = 2(1,1) + 2(1,0)$ and the same idea works: how much should we scale the vectors $(1,1)$ and $(1,0)$ so that we obtain $(4,2)$ by the parallelogram law of addition.

In the space $\mathbb{R}^3$, the vector $(4,2,3)$ can be written as

$$\begin{aligned}(4,2,3) &= 4(1,0,0) + 2(0,1,0) + 3(0,0,1)\\ &= 2(1,1,1) + 2(1,0,1/2) + 0(0,1,0)\end{aligned}$$

and again the same idea works, but this time in the space $\mathbb{R}^3$.

We formalize this concept as follows.

Definition of linear combination

Definition 1. Let V be a vector space and let $v_1, v_2, \cdots, v_n$ be vectors in V. A vector v in V of the form

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$$

where a_i are real numbers, is called a **linear combination** of the vectors $v_1, v_2, \cdots, v_n$.

Thus, we say $(4,2,3)$ is a linear combination of the vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ as well as the vectors $(1,1,1)$, $(1,0,1/2)$, $(0,1,0)$. Usually, we discuss linear combinations in the following two ways.

- Can the vector v be expressed as a linear combination of given vectors $v_1, \cdots, v_n$?
- What is the set of all possible linear combinations of the given vectors $v_1, \cdots, v_n$?

We discuss the first question with examples.

Example 1. The vector $(1,2)$ can never be expressed as a linear combination of $(1,0)$ and $(\pi,0)$. Can you think of a reason? Which are the vectors (a,b) that can be expressed as a linear combination of $(1,0)$ and $(\pi,0)$?

Example 2. Express, if possible, the vector $(0,3,0)$ as a linear combination of the vectors $(1,1,1)$, $(1,-1,1)$ and $(1,0,1)$ in $\mathbb{R}^3$.

In such cases, what is actually being asked is to, if possible, give the real coefficients $\alpha_1, \alpha_2, \alpha_3$ such that $(0,3,0) = \alpha_1(1,1,1)+\alpha_2(1,-1,1)+\alpha_3(1,0,1)$. If such $\alpha_i's$ were to exist, the right hand side simplifies and we get $(0,3,0) = (\alpha_1+\alpha_2+\alpha_3, \alpha_1-\alpha_2, \alpha_1+\alpha_2+\alpha_3)$. This gives us the system of equation

$$\begin{aligned}\alpha_1 + \alpha_2 + \alpha_3 &= 0\\ \alpha_1 - \alpha_2 &= 3\\ \alpha_1 + \alpha_2 + \alpha_3 &= 0\end{aligned}$$

which we solve by Gauss elimination or Gauss Jordan method to see if a solution exists. In the present example, we get the solution as $\alpha_1 = 2$, $\alpha_2 = -1$, $\alpha_3 = -1$. Thus, we have the linear combination $(0,3,0) = 2(1,1,1) + (-1)(1,-1,1) + (-1)(1,0,1)$.

Let us now look at a case, in which linear combination is not possible.

Example 3. Express, if possible, the vector $(0,3,0)$ as a linear combination of the vectors $(1,1,1)$, $(1,-1,-1)$ and $(1,0,0)$ in $\mathbb{R}^3$.

We proceed as before. We would like to obtain $\alpha_i's$ such that $(0,3,0) = \alpha_1(1,1,1) + \alpha_2(1,-1,-1) + \alpha_3(1,0,0)$. This gives us the system of equation

$$\begin{aligned} \alpha_1 + \alpha_2 + \alpha_3 &= 0 \\ \alpha_1 - \alpha_2 &= 3 \\ \alpha_1 - \alpha_2 &= 0 \end{aligned}$$

Gauss-elimination method gives

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 3 \\ 1 & -1 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 : R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 3 \\ 0 & 0 & 0 & -3 \end{pmatrix}$$

which shows that there is no solution (refer to notes of Module 1). Thus $(0,3,0)$ cannot be expressed as a linear combination of the given vectors $(1,1,1)$, $(1,-1,-1)$ and $(1,0,0)$.

Example 4. We are in the polynomial space $\mathcal{P}_3$ of all polynomials of degree ≤ 3. A typical element of this space is of the form $p(x) = a_1x^3 + a_2x^2 + a_1x + a_0$, where the $a_i's$ are real numbers. Then the vector $p(x)$ can be seen as a linear combination of $1, x, x^2, x^3$ which are vectors in $\mathcal{P}_3$.

Example 5. We are in the space $\mathcal{M}_2$ of all 2×2 matrices with real entries. Then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

is a linear combination in $\mathcal{M}_2$. Can you think of an analogous result in the space $\mathbb{R}^4$?

Note 1. The examples in the spaces of polynomials and matrices do not have the geometry of vector addition that is present in $\mathbb{R}^2$. What we have done above is the following: we converted the geometric idea of the parallelogram law of vector addition into an algebraic concept called linear combination and defined analogous ideas in case of polynomials and matrices.

Example 6. Express, if possible, the polynomial $v = t^2 + 4t - 3$ as a linear combination of the polynomials

$$u_1 = t^2 - 2t + 5, u_2 = 2t^2 - 3t, u_3 = t + 1$$

We set the equation

$$t^2 + 4t - 3 = \alpha_1(t^2 - 2t + 5) + \alpha_2(2t^2 - 3t) + \alpha_3(t + 1)$$

where our goal is to find the $\alpha_i's$. The method is to collect like powers of t in the right hand side of above equation and compare coefficients. Thus

$$t^2 + 4t - 3 = (\alpha_1 + 2\alpha_2)t^2 + (-2\alpha_1 - 3\alpha_2 + \alpha_3)t + (5\alpha_1 + \alpha_3).$$

We set the coefficients of like powers of t equal to each other to get a system of equation and use Gauss elimination method to obtain a solution, if it exists. Thus,

$$\alpha_1 + 2\alpha_2 = 1, \qquad -2\alpha_1 - 3\alpha_2 + \alpha_3 = 4, \qquad 5\alpha_1 + \alpha_3 = -3$$

which after Gauss elimination gives

$$\alpha_1 + 2\alpha_2 = 1, \qquad \alpha_2 + \alpha_3 = 6, \qquad 13\alpha_3 = 52$$

Back-substitution gives the solution $\alpha_1 = -3$, $\alpha_2 = 2$, $\alpha_3 = 4$. Thus $v = -3u_1 + 2u_2 + 4u_3$.

Example 7. Express, if possible, M as a linear combination of A, B, C where

$$M = \begin{pmatrix} 4 & 7 \\ 7 & 9 \end{pmatrix}, A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, C = \begin{pmatrix} 1 & 1 \\ 4 & 5 \end{pmatrix}$$

Just as we compared coefficients in case of polynomials in the above example, in case of matrices, we will write $M = xA + yB + zC$ and compare corresponding entries.

$$\begin{pmatrix} 4 & 7 \\ 7 & 9 \end{pmatrix} = x\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + y\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + z\begin{pmatrix} 1 & 1 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} x + y + z & x + 2y + z \\ x + 3y + 4z & x + 4y + 5z \end{pmatrix}$$

Comparing the corresponding entries, we get the system of equation

$$x + y + z = 4, \quad x + 2y + z = 7, \quad x + 3y + 4z = 7, \quad x + 4y + 5z = 9$$

Reducing to row echelon form gives

$$x + y + z = 4, \quad y = 3, \quad 3z = -3, \quad 4z = -4$$

The solution is $z = -1$, $y = 3$ and $x = 2$ and hence $M = 2A + 3B - C$.

Now, we discuss the second question for linear combinations: what can we say about the set of all possible linear combinations of a given set of vectors?

0.1 Span

Suppose that $\{v_1, v_2, \cdots, v_n\}$ is a set of n vectors in a vector space V. Collect all possible linear combinations of these vectors and form a set. This set is called the **span** of these vectors. That is

$$\text{Span}[v_1, \cdots, v_n] = \{\alpha_1 v_1 + \alpha_2 v_2 + \cdots \alpha_n v_n : \alpha_i \in \mathbb{R}\}.$$

Let us look at the geometry of some spanning sets.

Example 8. The span of the single vector $(1, 0)$ is by definition

$$\text{Span}[(1, 0)] = \{\alpha(1, 0) : \alpha \in \mathbb{R}\} = \{(\alpha, 0) : \alpha \in \mathbb{R}\}.$$

Can you identify this set as the $x - axis$? Can you think of a vector u whose span will be the $y - axis$?

Example 9. The span of the single vector $(1,1)$ is

$$\text{Span}[(1,1)] = \{\alpha(1,1)|\alpha \in \mathbb{R}\} = \{(\alpha,\alpha) : \alpha \in \mathbb{R}\}.$$

This set, geometrically is the line $y = x$. In general, the span of a vector u in $\mathbb{R}^2$ is the line passing through the vector u and the origin.

Example 10. The span of two vectors $(1,0)$ and $(0,1)$ is

$$\text{Span}[(1,0),(0,1)] = \{\alpha(1,0) + \beta(0,1) : \alpha,\beta \in \mathbb{R}\} = \{(\alpha,\beta) : \alpha,\beta \in \mathbb{R}\}$$

This is the set of all ordered pairs of real numbers and geometrically is the $\mathbb{R}^2$ space.

Similarly, write the span of the vectors $(1,1,0)$ and $(-1,0,0)$. Can you see it is the xy-plane in $\mathbb{R}^3$. Can you identify three vectors in $\mathbb{R}^3$ whose span is the entire $\mathbb{R}^3$?

Example 11. Consider the space $\mathcal{P}_2$. The span of $\{1,x,x^2\}$ is the set

$$\text{Span}[1,x,x^2] = \{a_0(1) + a_1(x) + a_2(x^2)|a_i \in \mathbb{R}\} = \{a_0 + a_1x + a_2x^2|a_i \in \mathbb{R}\}$$

which is actually the space of all polynomials of degree *at most* 2. We don't say exact degree 2 as a_2 can be zero. Similarly, Span$[x,x^2]$ will be subspace of all polynomials of degree at most two whose constant term is zero. Can you identify Span$[1,1+x,1+x^2]$ and describe any property of a polynomial (*coefficient, constant term, etc.*) that lies in the span?

Note 2. The idea of the set Span$[v_1,v_2,\cdots,v_n]$ is that if any vector u belongs to his set, then u is of the form $u = \alpha_1v_1 + \alpha_2v_2 + \cdots\alpha_nv_n$ and hence is a linear combination of $v_1,\cdots,v_n$. Hence the question of u being expressed as a linear combination of $v_1,v_2,\cdots,v_n$ can be re-framed as the question of whether u lies in the Span$[v_1,v_2,\cdots,v_n]$ and vice-versa. We have discussed above how to answer this.

0.2 Linear dependence

A set of vectors $\{v_1,v_2,\cdots,v_n\}$ is said to be **linearly dependent (LD)** if one of them can be expressed as a linear combination of others. That is, there exists a vector v_k in this set such that

$$v_k = \alpha_1v_1 + \alpha_2v_2 + \cdots + \alpha_{k-1}v_{k-1} + \alpha_{k+1}v_{k+1} + \cdots + \alpha_nv_n$$

We understand the above equality as the value of v_k being dependent on the values of other vectors in the set. Thus, in some sense, v_k does not really play any role in the set and can be safely omitted as its place is taken by the linear combination.

Example 12. The set $\{(1,0),(0,1),(1,1)\}$ is LD since $(1,1)$ is a linear combination of $(1,0)$ and $(0,1)$. Another example is the set $\{(1,1,1),(2,2,2)\}$ as $(2,2,2) = 2(1,1,1)$.

However, in general, it is difficult to identify the vector v_k which is a linear combination of others. In practice, we check for linear dependence as follows.

A set of vectors $\{v_1,v_2,\cdots,v_n\}$ is said to be linearly dependent if the equation

$$\alpha_1v_1 + \alpha_2v_2 + \cdots + \alpha_nv_n = \Theta$$

implies that at least one of the $\alpha_i's$ is non-zero. Here Θ is the additive identity of the underlying vector space V.

Example 13. Verify if the set of vectors $\{(1,4,2),(-2,2,-1),(-1,6,1),(0,8,3)\}$ is linearly dependent.

Since the vectors belong to $\mathbb{R}^3$, the additive identity is $\Theta = (0,0,0)$. Hence we form the equation $\alpha_1(1,4,2)+\alpha_2(-2,2,-1)+\alpha_3(-1,6,1)+\alpha_4(0,8,3) = (0,0,0)$ which gives the system of equations

$$\begin{aligned} \alpha_1 - 2\alpha_2 - \alpha_3 &= 0 \\ 4\alpha_1 + 2\alpha_2 + 6\alpha_3 + 8\alpha_4 &= 0 \\ 2\alpha_1 - \alpha_2 + \alpha_3 + 3\alpha_4 &= 0 \end{aligned}$$

We use Gauss elimination or Gauss Jordan method to obtain $\alpha_1 = 1$, $\alpha_2 = 1$, $\alpha_3 = -1$ and $\alpha_4 = 0$. Since at least one of the $\alpha_i's$ is not equal to zero (in fact there are three α_i's that are not equal to zero), the given set of vectors is linearly dependent.

0.3 Linear independence

A set of vectors $v_1, v_2, \cdots, v_n$ is said to be **linearly independent (LI)** if it is not linearly dependent. That is, none of the vectors can be expressed as a linear combination of other vectors. In practice, we check this as follows.

A set of vectors $\{v_1, v_2, \cdots, v_n\}$ *is linearly independent if the equation*

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = \Theta$$

implies that $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$.

Note 3. If you have an idea of statements and negations, you might be able to see that the definition of LI is the negation of the definition of LD: given a linear combination $\alpha_1 u_1 + \cdots \alpha_n u_n$, for LD we have $\exists \alpha_i (\alpha_i \neq 0)$ while for LI we have $\forall \alpha_i (\alpha_i = 0)$.

Example 14. Verify the linear dependence or linear independence of the set of vectors $\{(4,2,9),(0,1,3),(0,0,8)\}$.

We form the equation

$$\alpha_1(4,2,9) + \alpha_2(0,1,3) + \alpha_3(0,0,8) = (0,0,0)$$

This gives us the system of equation

$$4\alpha_1 = 0; \qquad 2\alpha_1 + \alpha_2 = 0; \qquad 9\alpha_1 + 3\alpha_2 + 8\alpha_3 = 0$$

which implies that $\alpha_1 = \alpha_2 = \alpha_3 = 0$. Hence the given set of vectors is linearly independent.

Example 15. Let us check the linear dependence or independence of the set of vectors $1+x, 1-x, x^2$ in $\mathcal{P}_2$. We form the linear combination

$$a_1(1+x) + a_2(1-x) + a_3x^2 = \Theta.$$

Since the underlying space is $\mathcal{P}^2$, Θ being the zero vector of $\mathcal{P}^2$ is the zero polynomial $0 + 0x + 0x^2$. Hence comparing coefficients, we get

$$a_1 + a_2 = 0; \qquad a_1 - a_2 = 0; \qquad a_3 = 0$$

which gives $a_1 = a_2 = a_3 = 0$. Thus the set is LI. What about the set of vectors $\{-3 + 2x + 2x^3, 1 - x^3, 2x - 1\}$? Again, we form the linear combination

$$a_1(-3 + 2x + 2x^3) + a_2(1 - x^3) + a_3(2x - 1) = \Theta = 0 + 0x + 0x^2 + 0x^3$$

This gives,

$$-3a_1 + a_2 - a_3 = 0, \qquad 2a_1 + 2a_3 = 0, \qquad 2a_1 - a_2 = 0$$

The solution is $a_1 = 1$, $a_2 = 2$, $a_3 = -1$. Since one of the a_i is non-zero, the set is LD.

Note 4. The working rules for LI and LD is seen as follows:

- Given a set of vectors $\{v_1, v_2, \cdots, v_n\}$, form the linear combination

  $$\alpha_1 v_1 + \cdots + \alpha_n v_n = \Theta$$

  and identify the zero vector Θ from the underlying vector space.
- This will reduce to a system of equations in $\alpha_i's$ which we solve by Gauss elimination method.
- If we get $(0, 0, \cdots, 0)$ as the only solution, that is, all the $\alpha_i's$ are zero, the set is LI. If we get *any* other solution, the set is LD.

Note 5. Let us consolidate the concepts of span of a set of vectors, linear dependence and independence.

- Given a set of vectors $\{v_1, v_2, \cdots, v_n\}$ there are vectors which can and which cannot be expressed as a linear combination of $v_1, v_2, \cdots, v_n$.
- If the vector v can be expressed as a linear combination of the $v_i's$, then v belongs to Span$[v_1, v_2, \cdots, v_n]$. If not, v does not belong to the span.
- If the set $\{v_1, v_2, \cdots, v_n\}$ is LD, there is a vector v_k in the set which can be omitted from Span$[v_1, v_2, \cdots, v_n]$ without losing anything. If the set is LI, all the $v_i's$ are equally important and none can be ignored.
- Till now, we have only discussed how to check whether a set is LD. In case it is, we have not seen how to identify the v_k which is a linear combination of the remaining vectors. Later, while discussing column space and rank, we will discuss a method to identify vectors which make the set LD.

The next concept is based on the following question. Suppose we are in a vector space or one of its subspace. *Can all the vectors of the vector space or subspace be expressed as a linear combination of some set of vectors* $\{v_1, v_2, \cdots, v_n\}$ *such that* n *is the* ***minimum*** *number of vectors satisfying this property?*

Exercises

1. In the following exercises, give an explicit description of span of S.
   (a) $S = \{(2,-1,-2),(1,3,-1)\}$, ($\{(x,y,z) \in \mathbb{R}^3 | x+z=0\}$)
   (b) $S = \{(1,1,2),(2,3,1),(1,2,-1)\}$ ($\{(x,y,z) \in \mathbb{R}^3 | z = 2x-3y\}$)
   (c) $S = \{x,(x+1)^2, x^2+3x+1\}$ ($\{ax^2+b+c | a=c\}$)
   (d) $S = \{x^2-4, 2-x, x^2+x+2\}$ (space of all polynomials of degree at most 2)
2. In the following exercises determine the span of S and whether S is a linearly dependent or independent set of vectors.
   (a) $S = \{(1,1,2),(0,-1,1),(2,5,1)\}$, (LD, $\{(x,y,z) \in \mathbb{R}^3 | z = 3x-y\}$)
   (b) $S = \{(3,3,2),(0,0,0),(1,-1,-1)\}$ (LD, $\{(x,y,z) \in \mathbb{R}^3 | x-5y+6z=0\}$)
   (c) $S = \{(1,2,1),(-1,0,3),(0,1,1),(2,1,1)\}$ (LD, $\mathbb{R}^3$)
3. Let $S = \{1, x-3, x^2+2x, 2x^2+3x+5\}$.
   - Find the span of S.
   - Is S linearly independent?
   - Show that $2x^2+3x+5$ is a linear combination of the other vectors in S.
4. Let $T = \{1, x-3, x^2+2x, x^3\}$.
   - Is T linearly independent?
   - Is span of T equal to $\mathcal{P}_3$?
5. Let $S_1 = \{(2s-t, s, t, -s) | s,t \in \mathbb{R}\}$ and $S_2 = \{(-s, s-5t, 3t+2s) | s,t \in \mathbb{R}\}$.
   - Show that S_1 and S_2 are subspaces of $\mathbb{R}^4$ and $\mathbb{R}^3$ respectively.
   - Find a set of vectors that span S_1 and a set of vectors that span S_2.
   - Is span of S_1 equal to $\mathbb{R}^4$ and span of S_2 equal to $\mathbb{R}^3$?